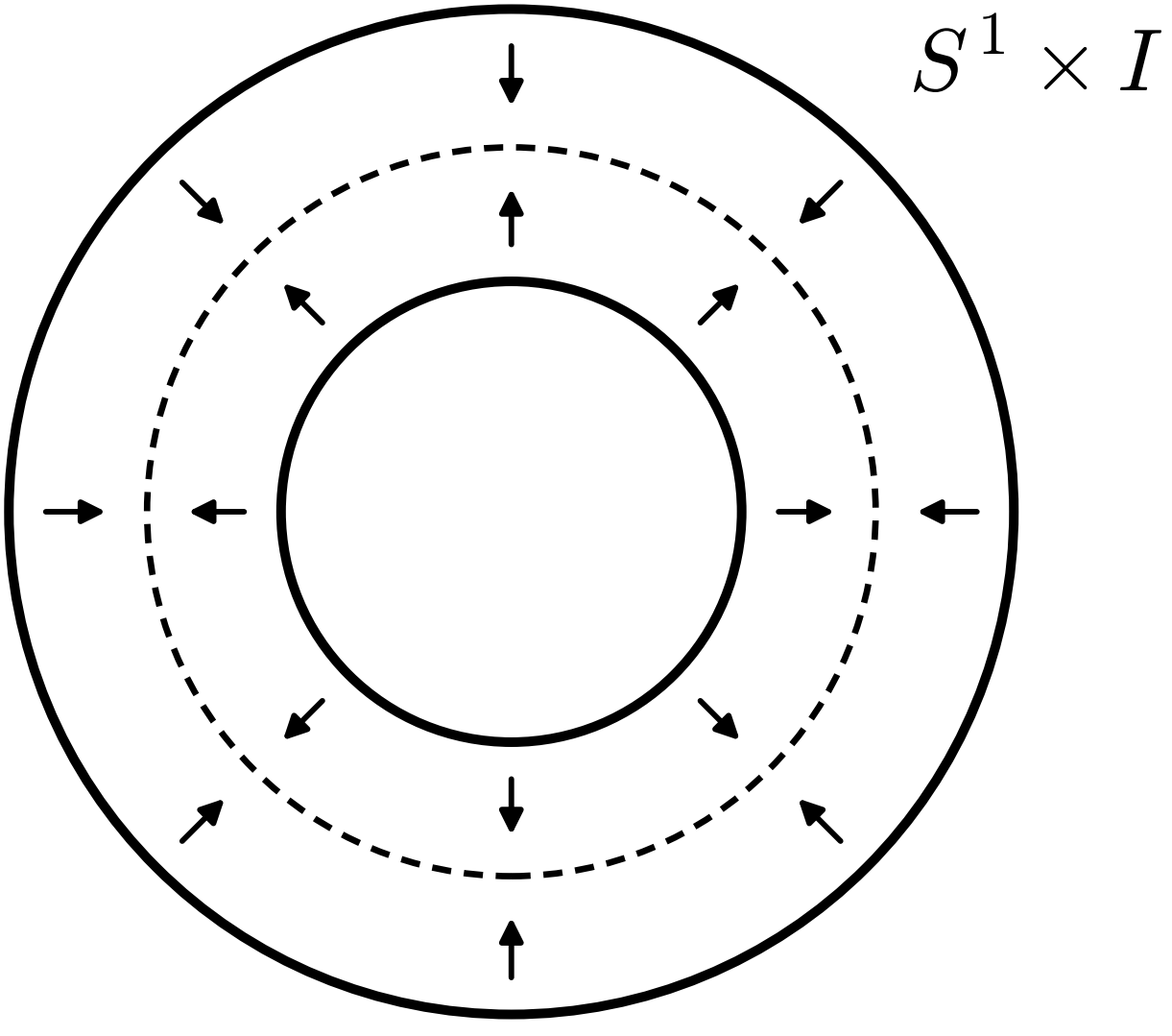
$S^1 \times I$